FACULTY OF ENGINEERING & TECHNOLOGY

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT

INTERFACING TECHNIQUES ENCS4380

Mechanical Components and Materials



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Mechanical design proposal

This report recommends the mechanical components and materials for the ENCS4380 Micromouse robot. The proposed design uses an original printed PLA chassis, two N20 encoder motors, rubber tired wheels and a compact caster.

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Construction

Use a 2–2.5 mm base with ribs, reinforced screw bosses and rounded edges. Print the motor clamps, five sensor holders and a removable battery tray with the chassis or as replaceable parts. Start with a tolerance test before committing to the final CAD.

Material	Local price or cost basis	Design decision
PLA	Azr CCF PLA: ₪80 for 1 kg [3]	Preferred first chassis. Rigid and easy to prototype; reinforce screw bosses and avoid thin clips.
PETG	Azr PETG options from ₪85; confirm pack size [3]	Alternative for durable clamps or final chassis. More flexible; print and fit test before use.

PLA is recommended for the first build because it is rigid and straightforward to print. PETG is a useful option for tougher replacement clamps. Sheet materials can support early mock-ups or reinforcement, but do not replace the required original printed chassis.

Supplier comparison

RoboticX offers listed encoder motors and a small caster. Azr Market lists PLA and PETG locally. Campus printing is the recommended fabrication route, subject to booking and fees. Generic fasteners require a local shop quotation. No compatible standalone local wheel listing has been confirmed.



Proposed components and local sourcing

Prices are in shekels, checked on 1 November 2026. Allowances require local quotations. Supplier links are on page 6.

Component and specification	Qty	Local source	Price basis	Line total
N20 encoder motor, 6 V, 530 RPM; verify shaft and encoder dimensions	2	RoboticX [1]	Listed ₪75 each	₪150
Small steel ball caster with plates, retaining ring and screws	1	RoboticX [2]	Listed	₪8
PLA filament, 1.75 mm; 60–100 g for chassis and printed mounts	60–100 g	Azr Market [3]	₪80 per kg; consumed material	₪4.80–8
Rubber tired drive wheels, preferably 32–34 mm diameter; match actual motor shaft	2	Request from RoboticX	Standalone supply and price unconfirmed	Quote
Screw retained N20 motor clamps	2	Campus 3D printers	Included in PLA above	Included
Sensor holders: front, two diagonal, two side	5	Campus 3D printers	Included in PLA above	Included
Battery tray with rounded edges and strap slots	1	Campus 3D printers	Included in PLA above	Included
Controller supports, printed spacers or matching nylon standoffs	4–8	Campus / local hardware shop*	Allowance	₪5–15
M2/M3 fasteners; 20–30 screws plus nuts and washers; size after CAD	1 lot	Local hardware shop*	Allowance	₪10–20
Hook and loop battery strap plus thin EVA pad	1 set	Local sewing / hardware shop*	Allowance	₪5–10
Small cable ties and heat shrink	1 lot	Local electronics shop*	Allowance	₪5–10

Subtotal: ₪187.80–221, excluding wheels, printing fees, delivery, battery and electronics. Generic hardware suppliers marked * require stock confirmation and quotations.



Dimensions and mounting requirements

Project requirement	Proposed design and verification
Each dimension under 200 mm	Target 110 × 100 × about 60 mm including wheels, sensors and wires. Measure the finished robot.
Turn within a nominal 168 mm corridor with 5% tolerance	Use 159.6 mm as a conservative clear width. A centered 110 × 100 mm rectangle sweeps a 148.7 mm diameter; verify the actual rotation center and protrusions in CAD.
3–5 mm ground clearance	Target 4 mm. Recess screws and route cables above the underside; test taped seams.
Sensors below 50 mm and aimed at walls	Place optical centers about 25–30 mm above floor. Keep side sensor faces about 80 mm apart for approximately 39.8 mm wall distance when centered in a 159.6 mm corridor.
Original printed chassis and total robot cost below $200	Emboss team name. Maintain a complete BOM including electronics and donated parts at market value.
Mass and race configuration	No numerical maximum mass appears in the supplied rules. Use 250 g as an internal ceiling. Do not change robot weight between mapping and racing.

Motor mounting

Drive system Align both motor shafts on one transverse axis near the footprint center. Aim for 85–90 mm wheel center spacing, subject to actual motor length, encoder boards and wheel hubs. Bolt the clamps down and leave clearance for rotating parts.

Battery and controller mounting

Battery and controller Place the battery low, close to the axle and slightly toward the caster. Retain it with a padded tray and strap so it can be swapped quickly. Mount boards on insulated supports with USB, reset and power controls accessible.

Sensor and caster mounting

Sensors and caster Use rigid, removable sensor mounts at front, ±45 degrees and sides. Keep optical fields clear. Measure the local caster before designing a raised mount or pocket so the chassis sits level at 4 mm clearance. Check that its steel ball does not mark the maze floor.



Mass budget and purchasing checks

Assembly	Estimated mass
Printed chassis, holders and tray	45–65 g
Two motors with encoders	25–40 g
Wheels, clamps and caster	15–25 g
Battery	20–35 g
Controller, sensors, driver and wiring	35–55 g
Fasteners and retention	10–20 g
Total planning range	150–240 g

These are estimates, not measured weights. The design target is 180–250 g assembled; 250 g is an internal ceiling, not a course limit. Keep the battery low and near the wheel axle, slightly toward the caster. Check that the center of mass remains within the wheel and caster support triangle during motion.

Mechanical budget

Using only consumed PLA gives a mechanical subtotal of ₪187.80–221. If the team must buy a whole ₪80 spool, the cash outlay becomes ₪263–293. Both figures exclude wheels, printer service, delivery, battery and electronics. Do not add the whole spool and consumed filament together.

Record actual purchases and use the applicable payment exchange rate when checking the complete robot against the $200 limit. Ask the instructor how shared filament and fabrication costs should be recorded. Final budget compliance cannot be confirmed until the wheel quote and electronics BOM are complete.



Purchasing checks and supplier references

Wheel selection

Confirm a pair of rubber tired wheels that fits the purchased motor shafts before freezing CAD. Prefer 32–34 mm diameter. RoboticX lists a ₪250 miniD kit containing 42 mm wheels, but separate wheel supply is unconfirmed. Ask for a standalone quote. Recheck mounting height and turning clearance if using 42 mm wheels. The supplied kit chassis cannot replace your required original printed chassis.

Construction checks

Print a hole and clamp tolerance sample. Start with 0.2 mm layers, three perimeters and 15–20% infill; reinforce bosses locally. Keep the first chassis print near three hours. Leave encoder magnets, wheel hubs and sensor fields unobstructed. Secure wires above the clearance plane.

Before final assembly

Verify motor shaft fit, wheel grip and caster height. Check the full turning envelope, floor seams, sensor aim and battery access. Measure the finished size and mass, and replace every allowance with a quotation or receipt. Order parts by 4 November; the mounted printed chassis is due 2 December 2026.

Supplier references

[1] RoboticX encoder motor
https://roboticx.ps/product/n20-dc-motor-with-magnetic-encoder-6v-530-rpm/

[2] RoboticX small caster
https://roboticx.ps/product/small-ball-caster-wheel/

[3] Azr Market filament
https://azrmarket.ps/en/c/filament

[4] RoboticX miniD kit
https://roboticx.ps/product/2wd-minid-robot-chassis/

Requirements: Micromouse Project Brief ENCS4380, 20 October 2026, pages 2–4 and 7; Milestone Plan, pages 3–4. Listings establish advertised prices, not reserved stock or guaranteed delivery.

